

# Lecture 1 – Projects, Processes, and Methods

---

1DT308/1DT902 Introducerande projekt

2025-05-13

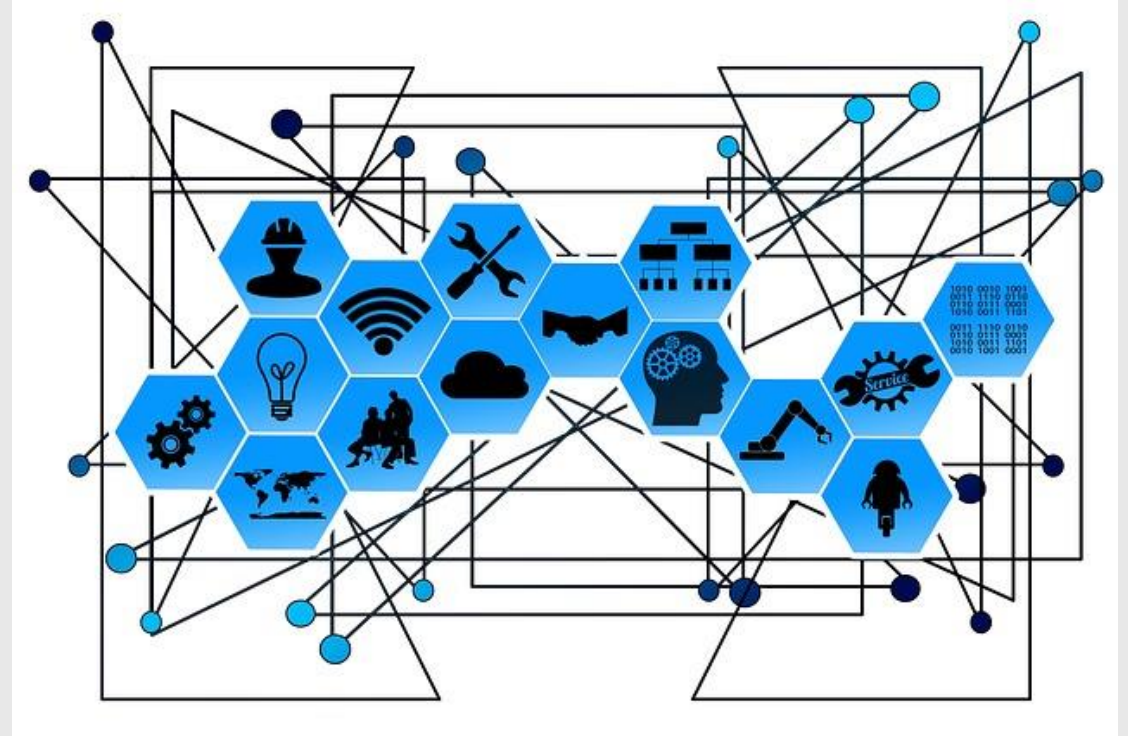
# Lectures

1. Projects and processes/methods
2. Some personal experiences

# Today's takeaways

- What is a project?
- Managing
- Some processes
- People

# Projects



# Project or something ongoing?

| Project                                            | Ongoing                                                   |
|----------------------------------------------------|-----------------------------------------------------------|
| Has a beginning and an end                         | No definitive beginning and end                           |
| Temporary in nature                                | Ongoing                                                   |
| Produces a unique product, service, or result      | Repeatingly produces the same product, service, or result |
| Resources are limited and dedicated to the project | Resources are dedicated to operations                     |
| Ending is determined by specific criteria          | Processes are ongoing                                     |

”A project is a temporary endeavor to accomplish a unique product, service or result”

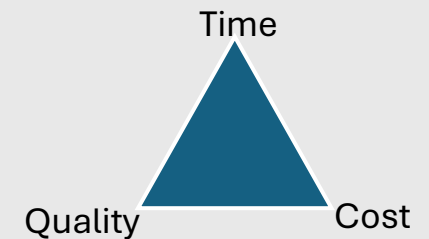
# Project life cycle



The series of phases that a project passes through from its start to its completion.

# Project management

- Project management is the discipline of achieving the project goals while adhering to the constraints
- Some aspects of project management:
  - establishment of clear and achievable goals
  - planning, organizing, securing and managing resources
  - balancing demands: scope, quality, time and cost
  - communication
  - challenges: human beings, changing/incomplete requirements



# Process groups (pmbok)

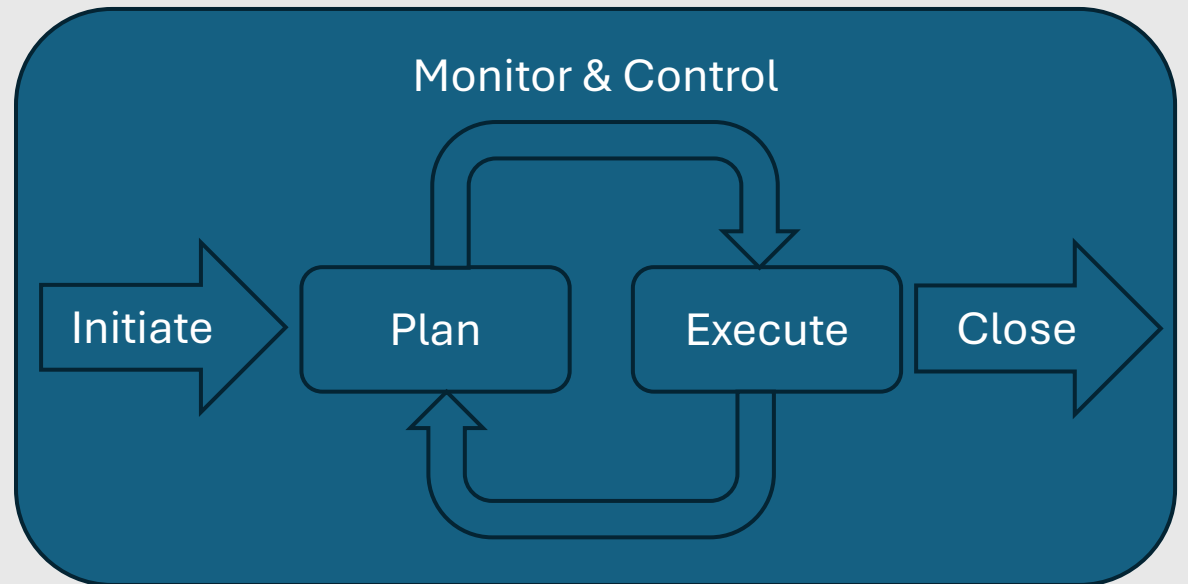
**Initiating:** Define the business need, set high-level goals and budgets, and start identifying key stakeholders.

**Planning:** Outline the project scope, schedule, resources, and risks, and plan for communications, quality, and procurement.

**Executing:** Assemble and manage the project team, engage with stakeholders, and carry out the project plans.

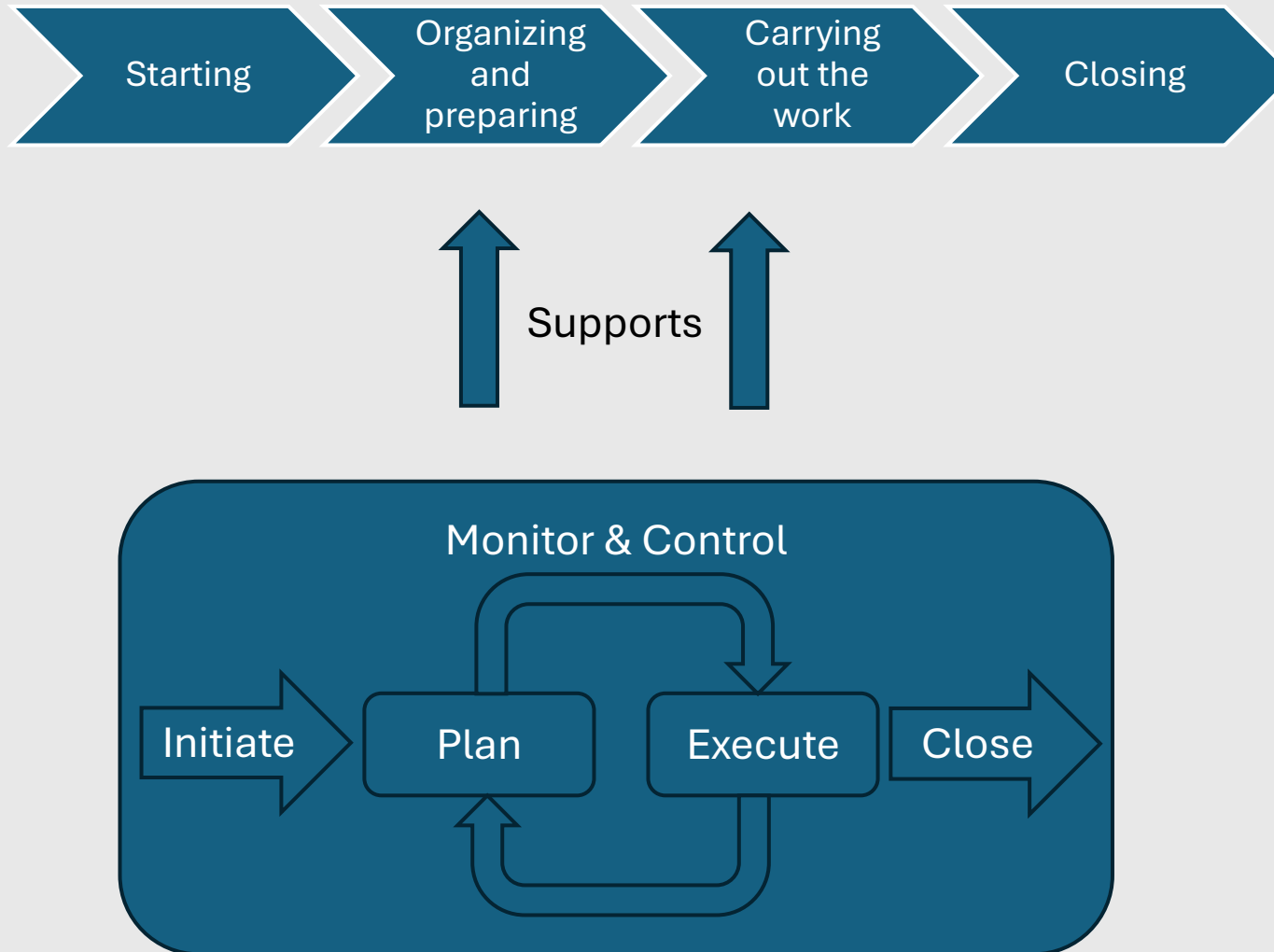
**Monitoring and Controlling:** Track progress and make necessary adjustments to ensure the project stays on track and meets its objectives.

**Closing:** Conclude all project activities and formally close the project.





# Life Cycle and Process Groups



# Software Processes

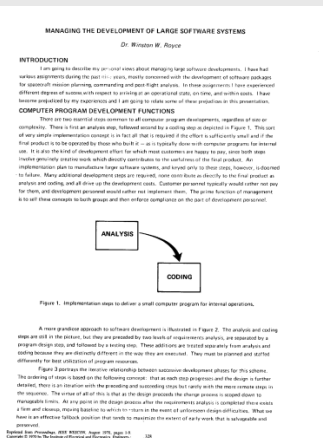
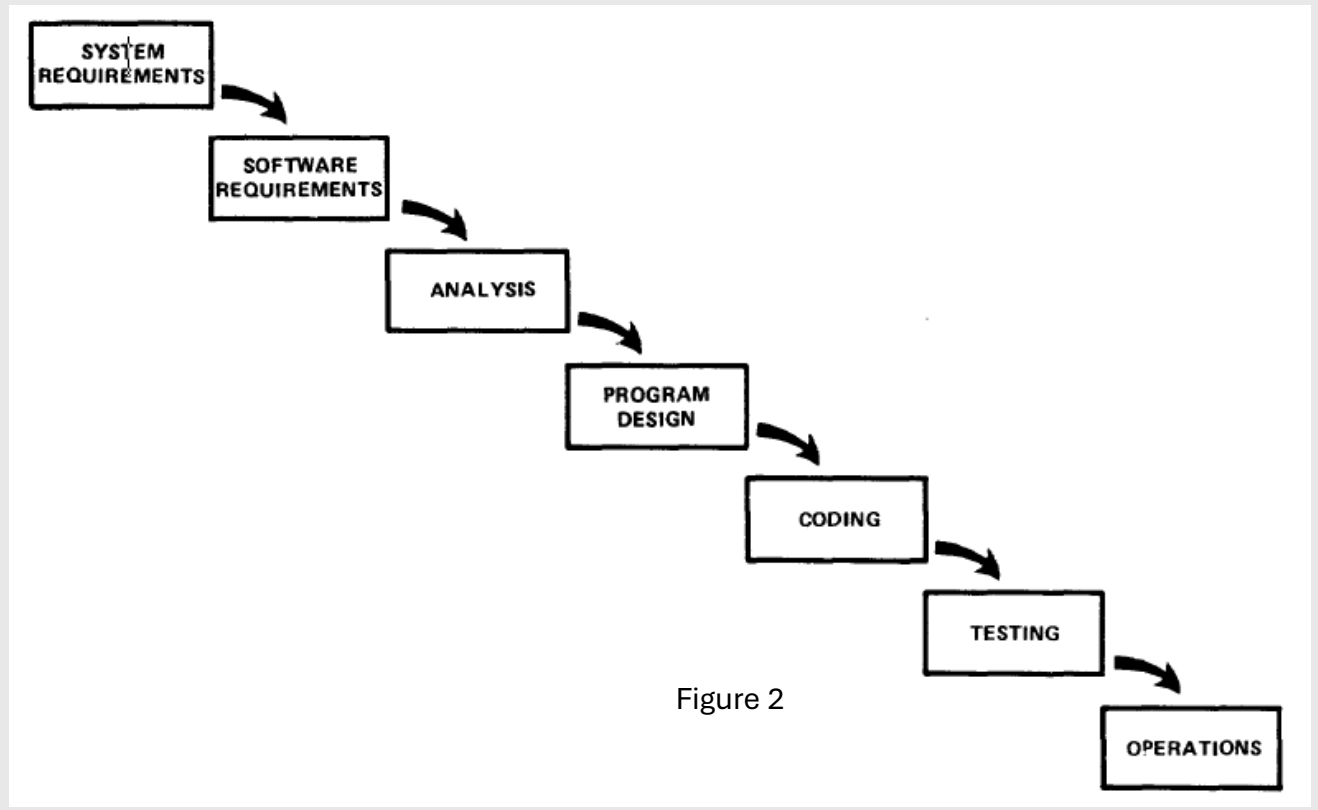
# Waterfall

## Managing The Development Of Large Software Systems

Dr Winston W. Royce, 1970

Sequential

Many have gone mental bezerk.

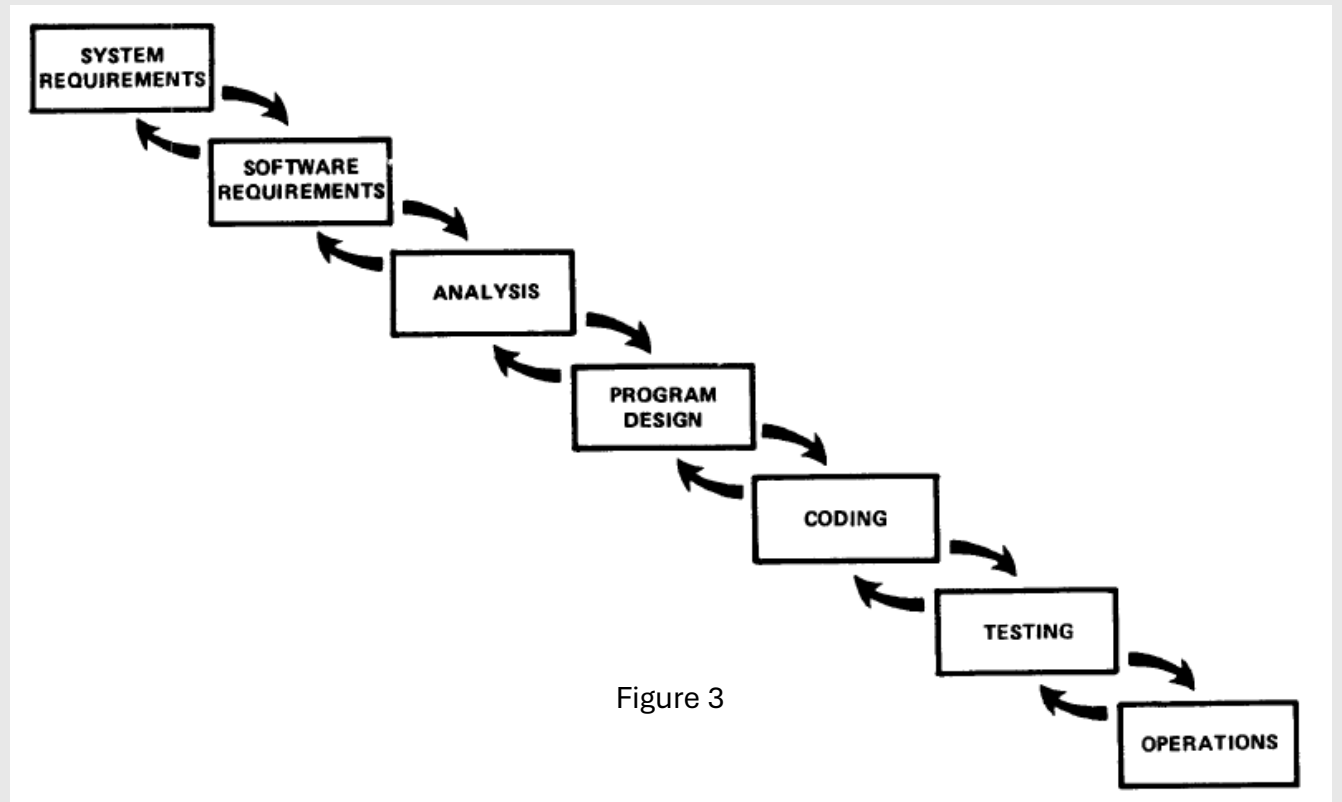


”An implementation plan to manufacture larger software systems, and keyed only to these steps, however, is doomed to failure.” – Page 1.

# Waterfall 2

Figure 3 portrays the **iterative relationship between successive development phases** for this scheme.

The ordering of steps is based on the following concept: **that as each step progresses and the design is further detailed, there is an iteration with the preceding and succeeding steps** but rarely with the more remote steps in the sequence.



## Waterfall 3

“I believe in this concept, but the implementation described above is risky and invites failure. The problem is illustrated in Figure 4. The testing phase which occurs at the end of the development cycle is the first event for which timing, storage, input/output transfers, etc., are experienced as distinguished from analyzed.”

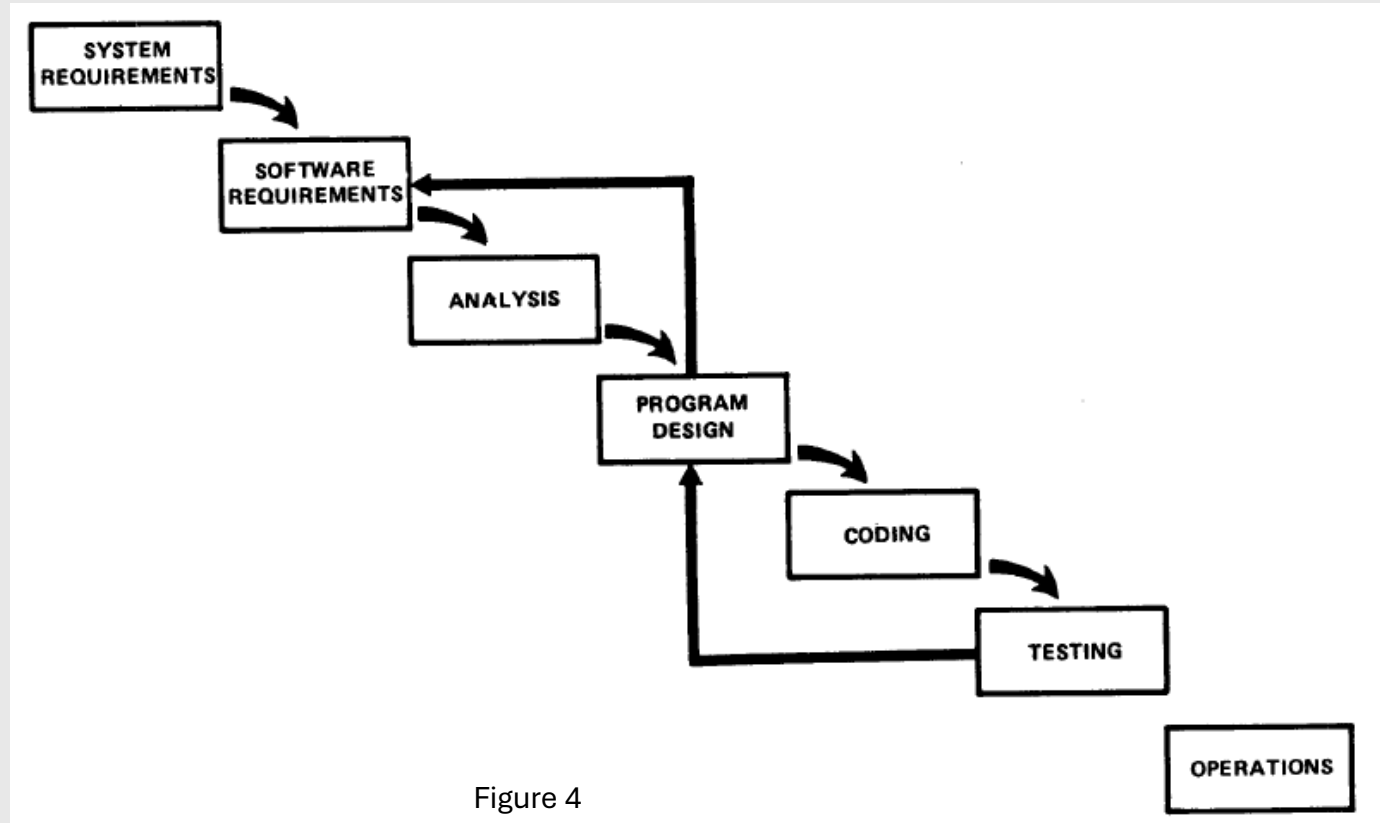


Figure 4

# Problem with traditional methodologies

Assumes a predictable world:

Clear requirements are known at start

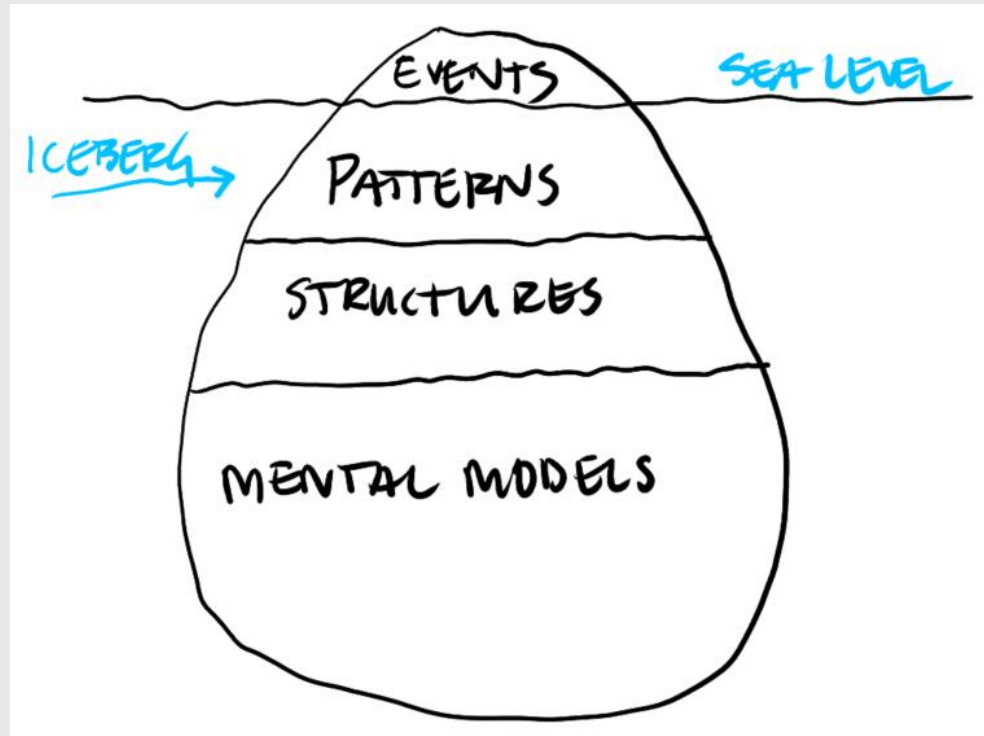
Competitive landscape doesn't change

Stakeholders don't change their minds

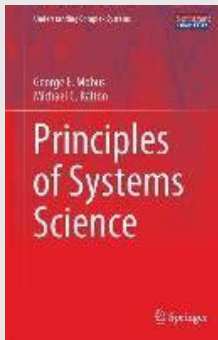
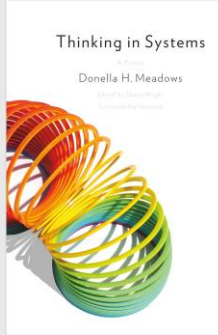
Availability of resources doesn't change



# Systems Thinking – Iceberg Model



Understanding complex problems by viewing them as systems affecting each other (i.e most likely non-linear in their behaviour).



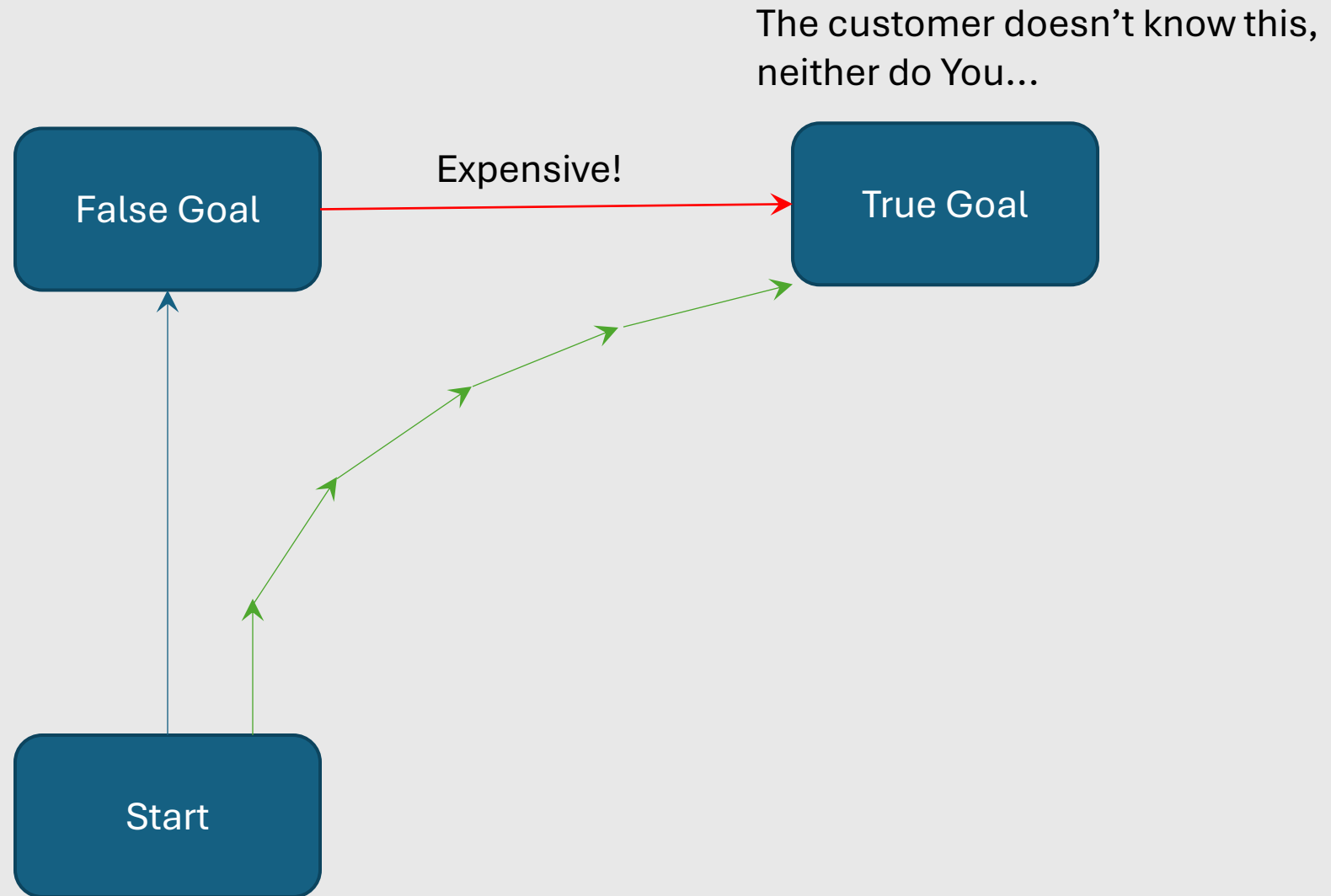
# How do you eat an elephant?



One bite at a time!



# How Do One Create Stuff In Real Life ? (Or not)



# Agile (meaning)

## The Free Dictionary

Characterized by quickness, lightness, and ease of movement; nimble.

## Merriam-Webster

- 1) marked by ready ability to move with quick easy grace
- 2) having a quick resourceful and adaptable character



# Agile Software Development

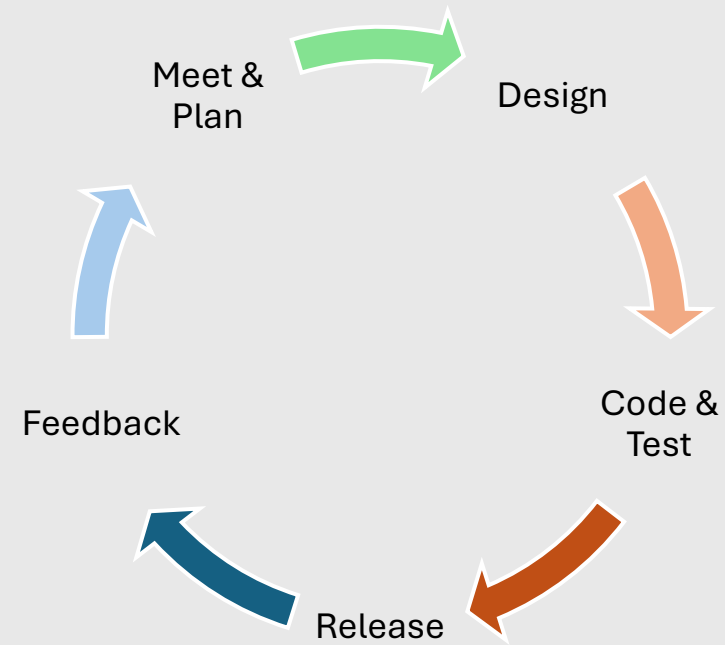
Iterations and increments

Value-centric

Frameworks: Scrum, Kanban, Extreme Programming, etc

Practices: Pair-programming, TDD, sprints, etc

<https://www.agilealliance.org/agile101/>



# Manifesto for Agile Software Development

We are uncovering better ways of developing software by doing it and helping others do it.  
Through this work we have come to value:

Individuals and interactions over processes and tools  
Working software over comprehensive documentation  
Customer collaboration over contract negotiation  
Responding to change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.



*We follow these principles:*

Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.

Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.

Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.

Business people and developers must work together daily throughout the project.

Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.

The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.

Working software is the primary measure of progress.

Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.

Continuous attention to technical excellence and good design enhances agility.

Simplicity--the art of maximizing the amount of work not done--is essential.

The best architectures, requirements, and designs emerge from self-organizing teams.

At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

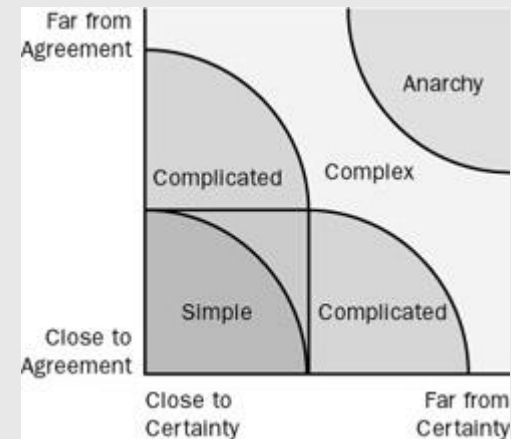
# Scrum

Scrum is an agile process that allows us to focus on delivering the highest **business value** in the shortest time.

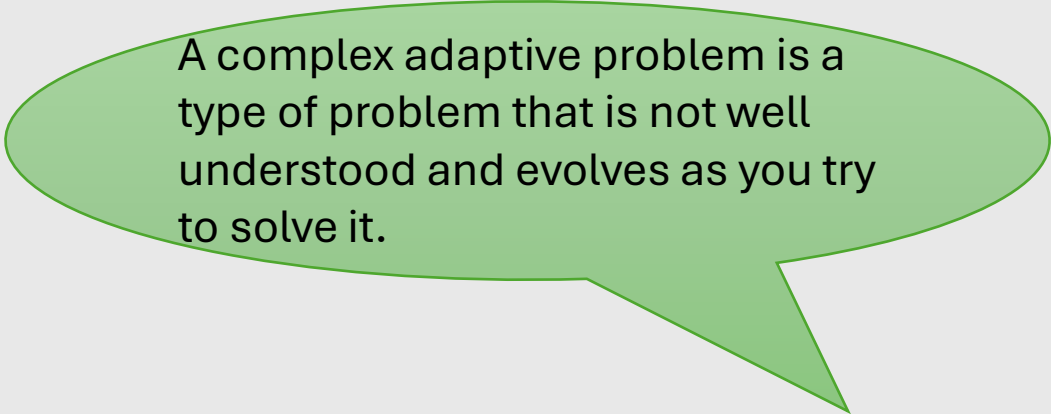
It allows us to rapidly and repeatedly inspect actual working software (every two weeks to one month).

The business sets the priorities. Teams self-organize to determine the best way to deliver the highest priority features.

Every two weeks to a month anyone can see real working software and decide to release it as is or continue to enhance it for another sprint.



# Definition of Scrum



A complex adaptive problem is a type of problem that is not well understood and evolves as you try to solve it.

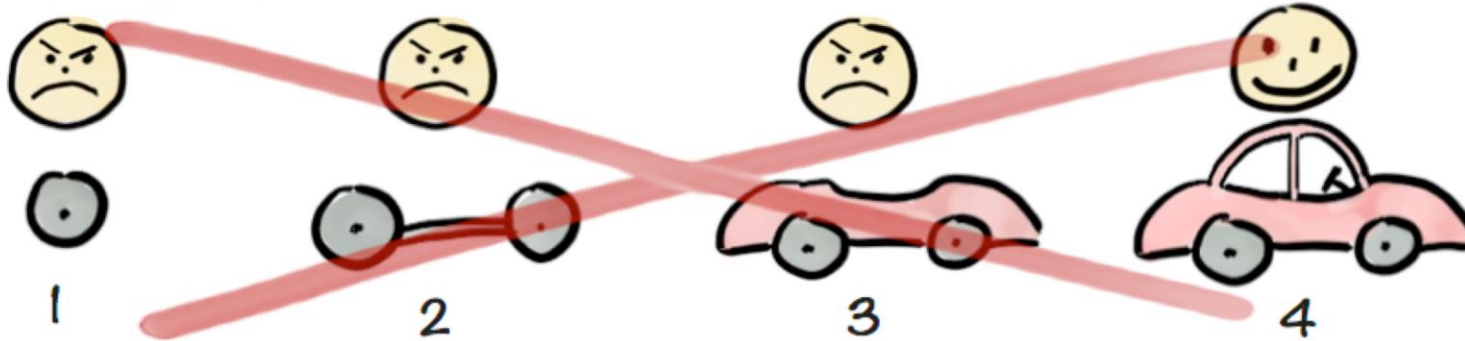
Scrum (n): A **framework** within which people can address **complex adaptive problems**, while productively and creatively delivering products of the **highest possible value**.

Lightweight - Simple to understand - Difficult to master

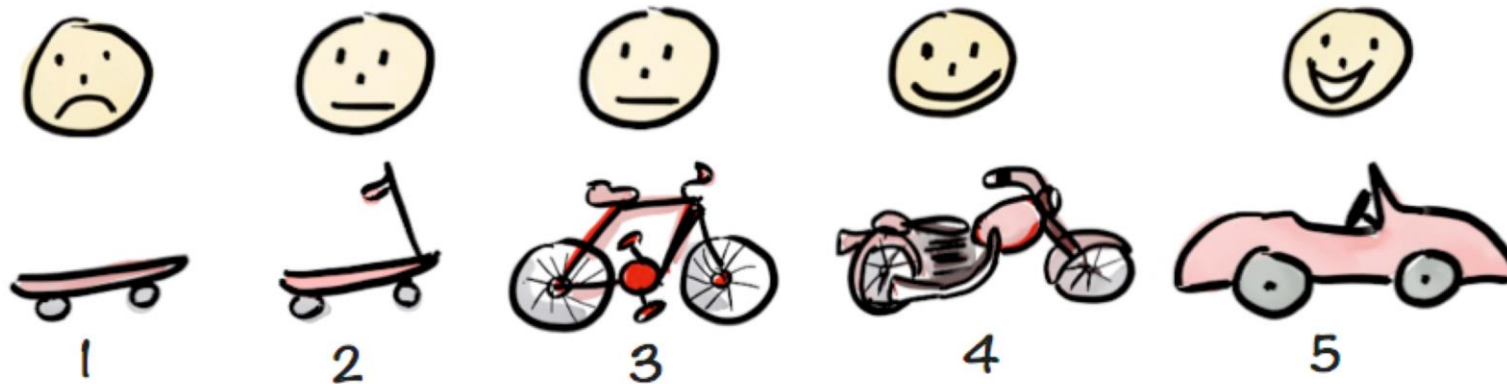


# Scrum vs. Traditional

Not like this....



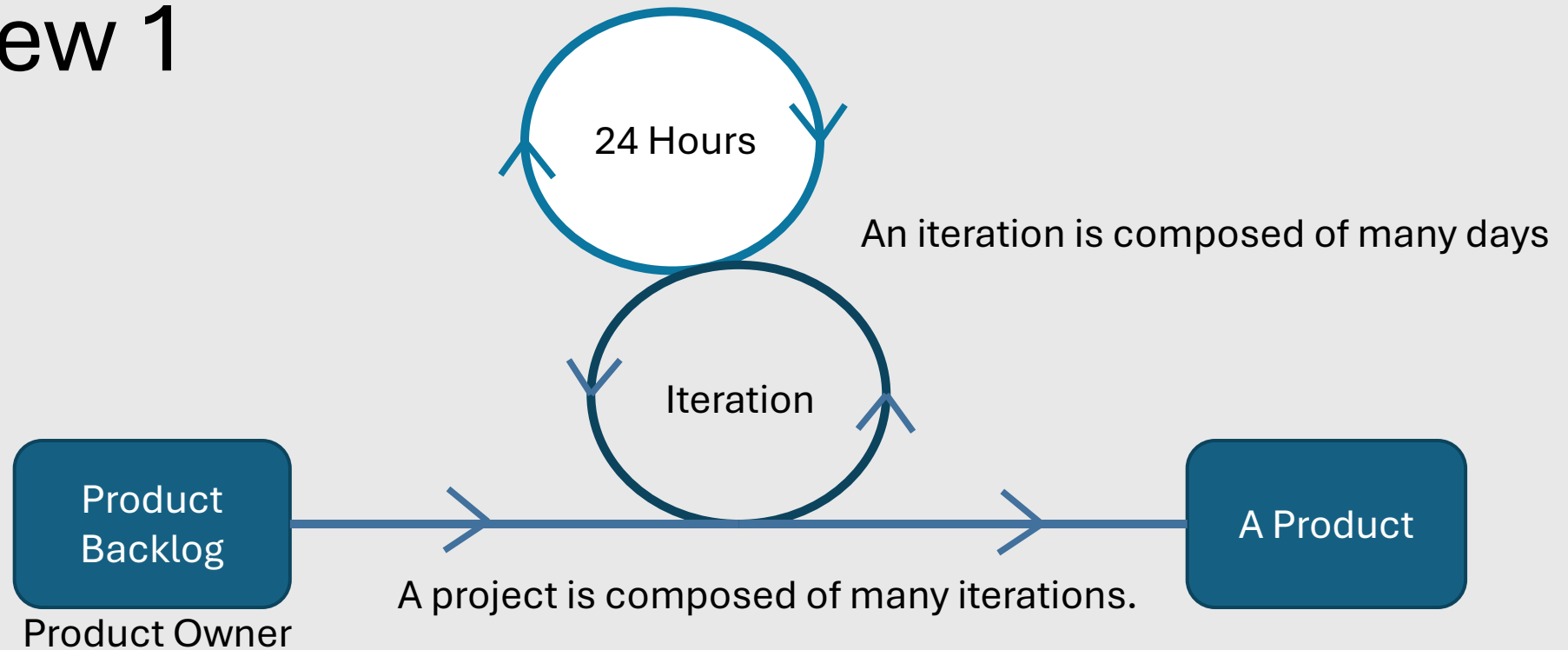
Like this!



by Henrik Kniberg

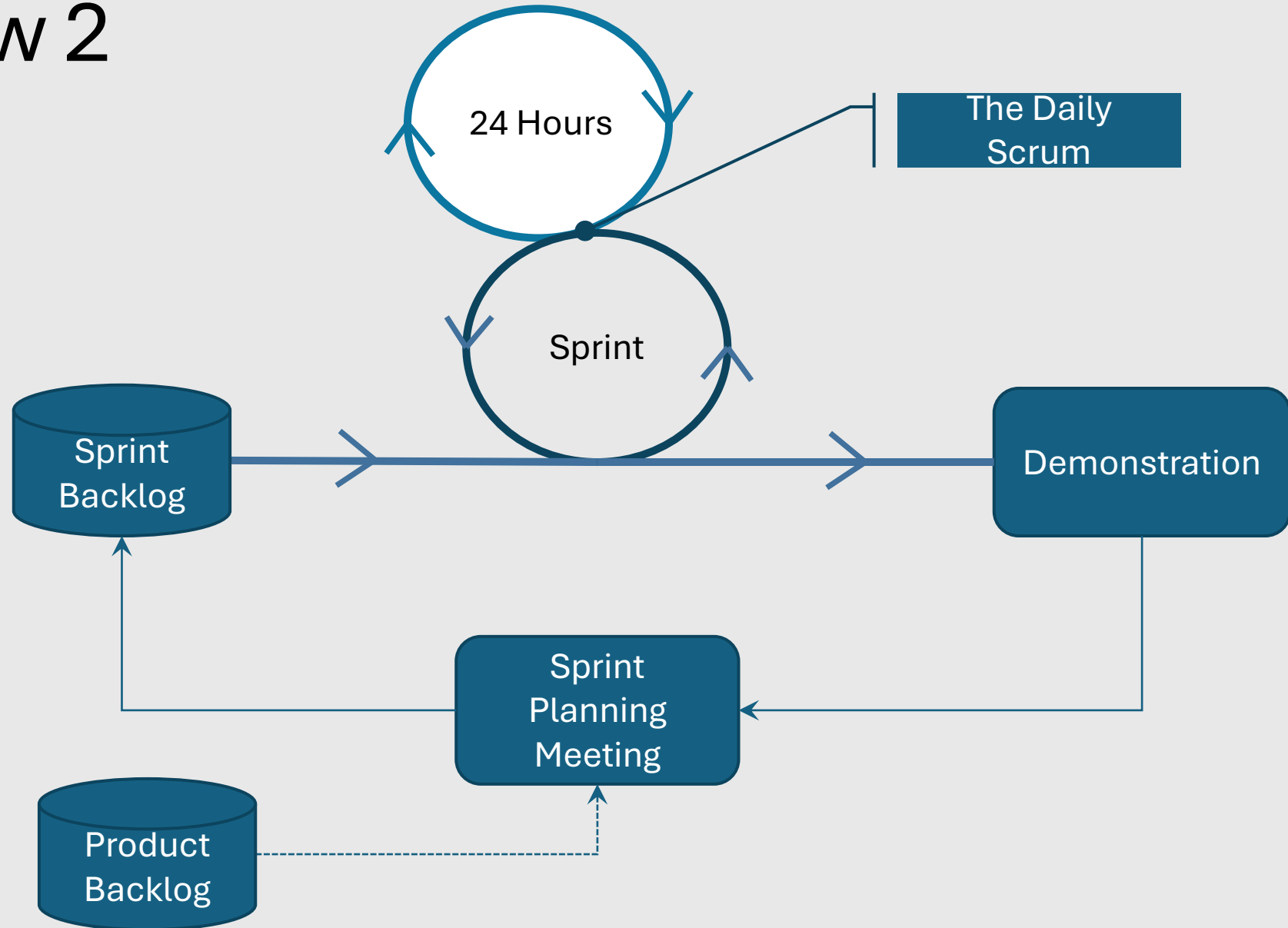


# Overview 1

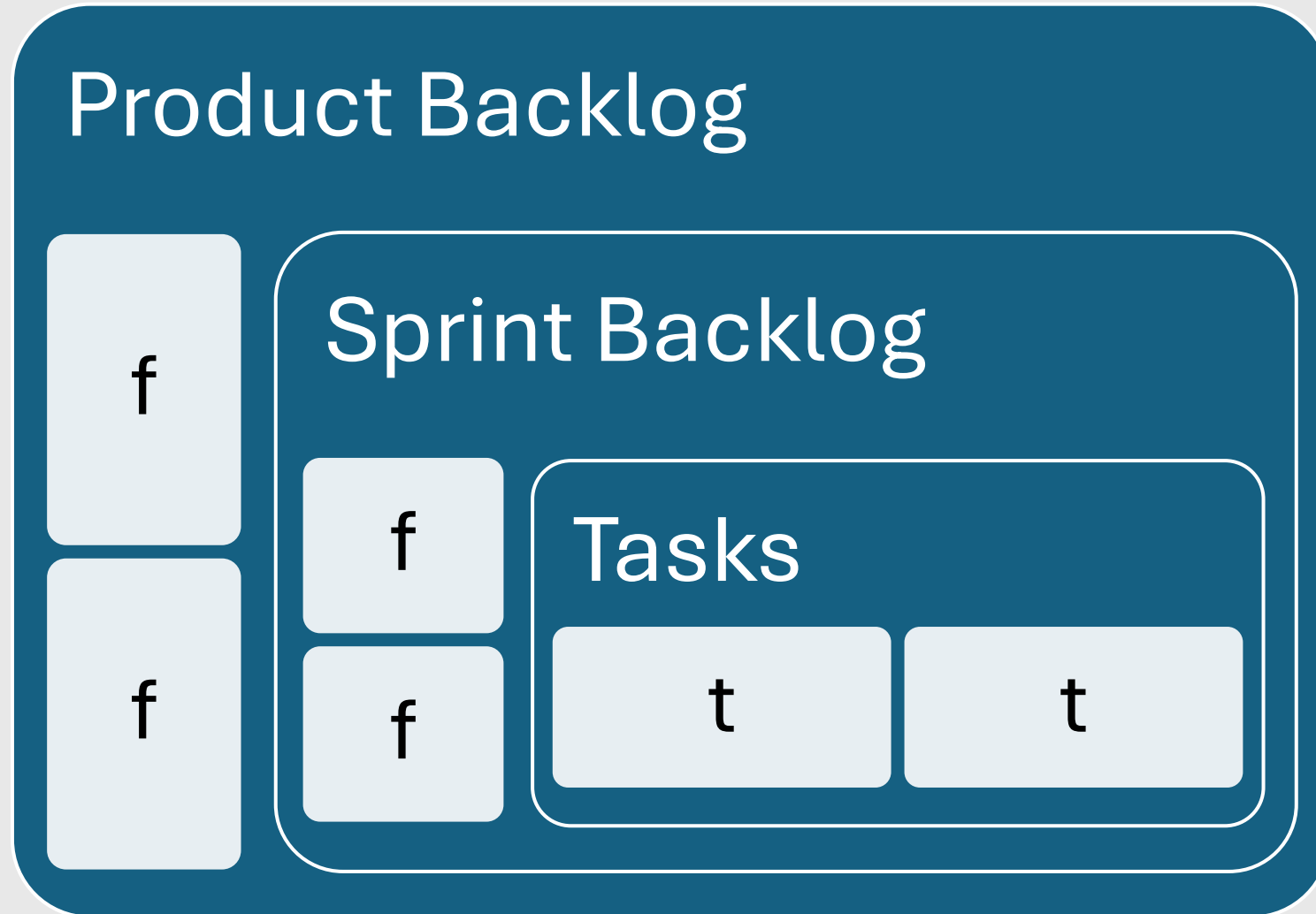


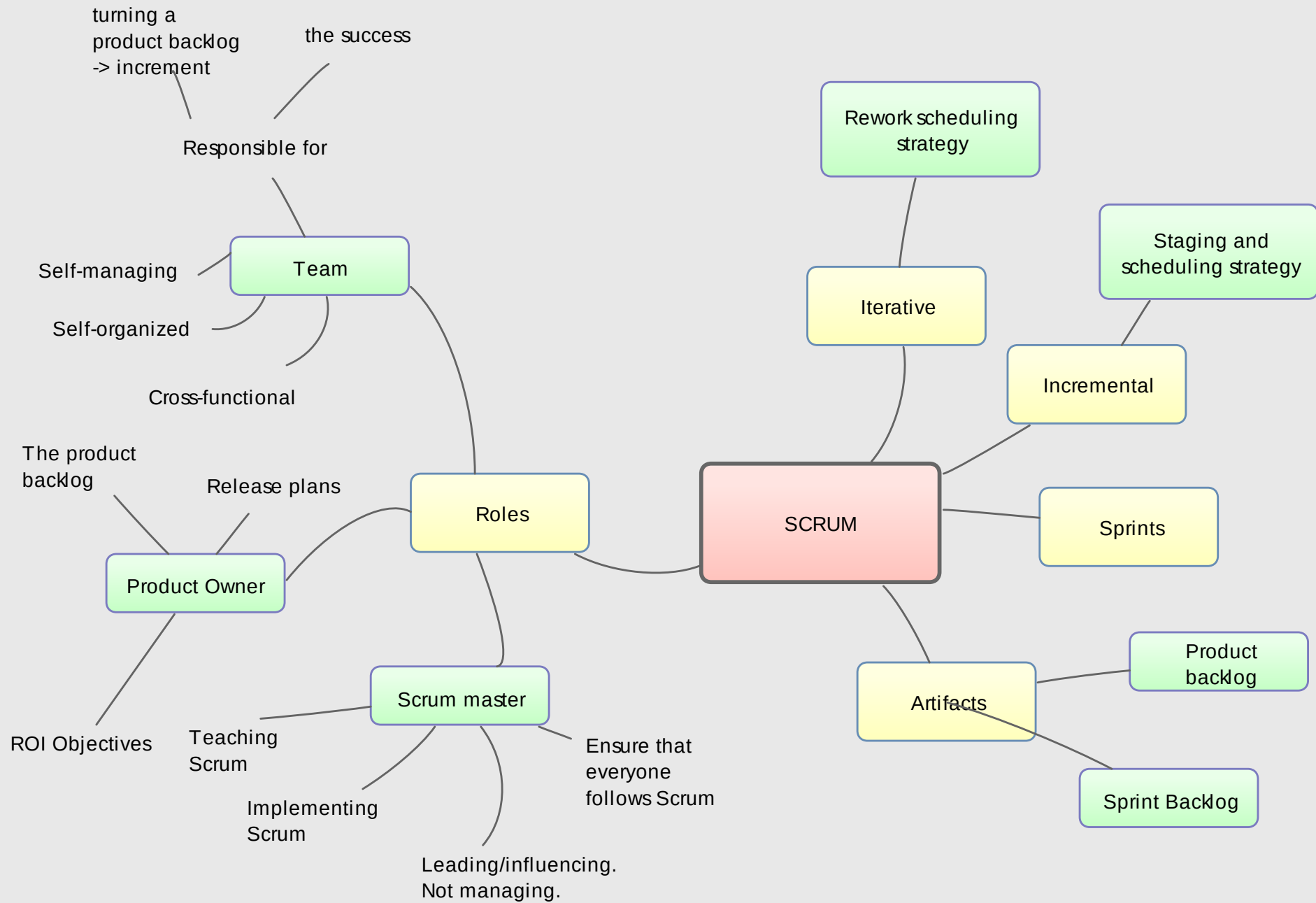
Iterations and Increments!

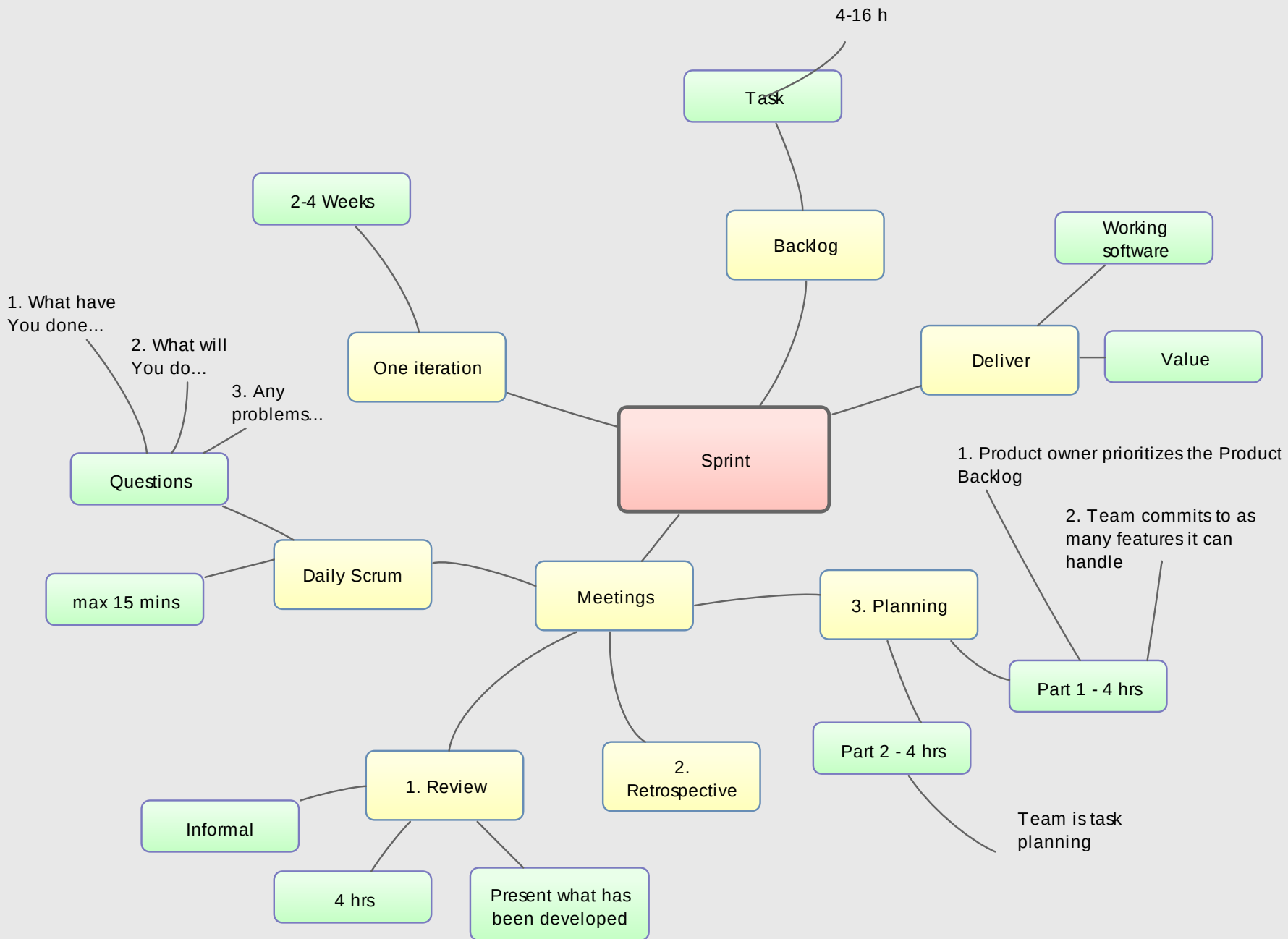
# Overview 2

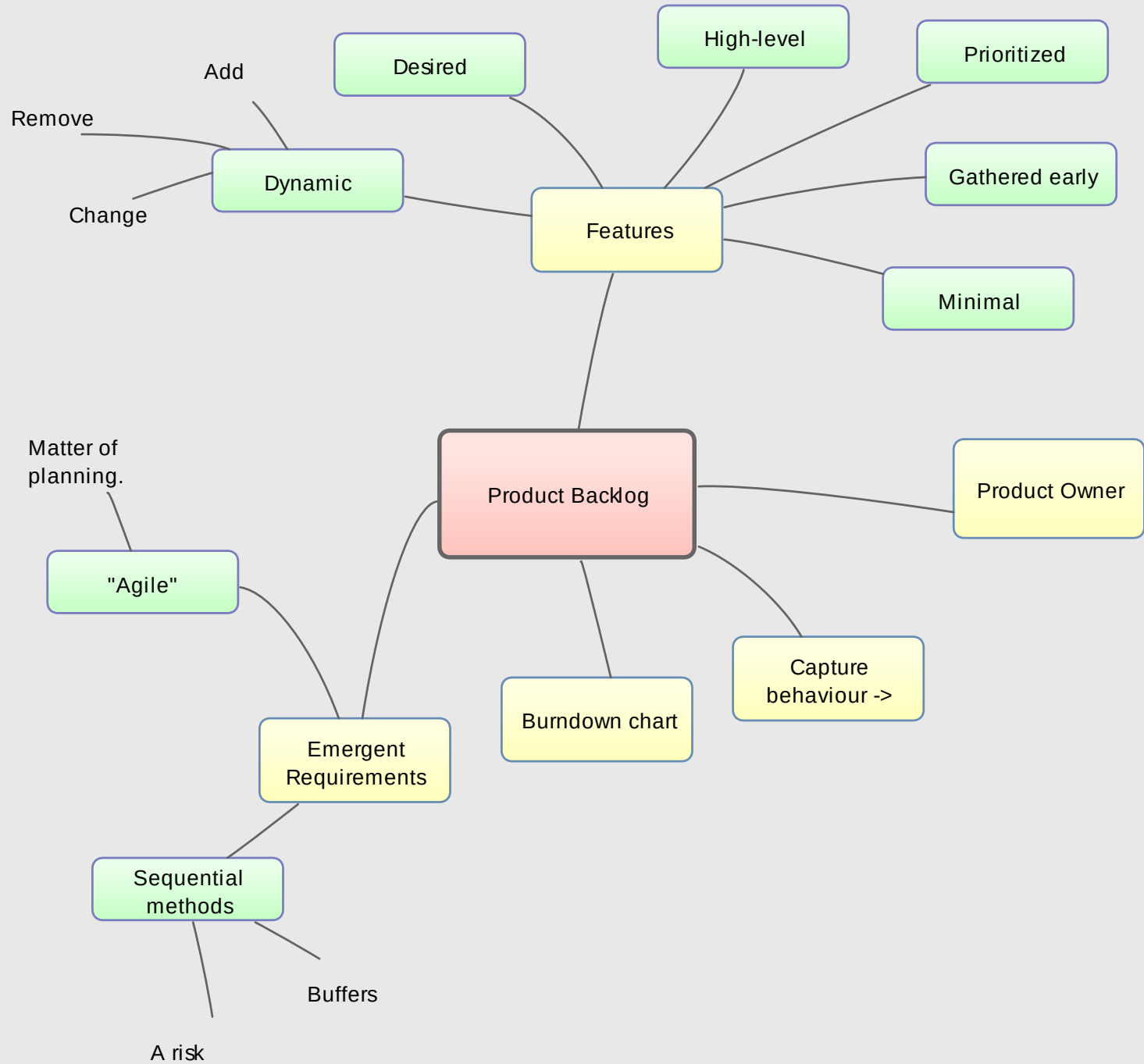


# Backlogs & Tasks









# Unified Process

A software engineering process

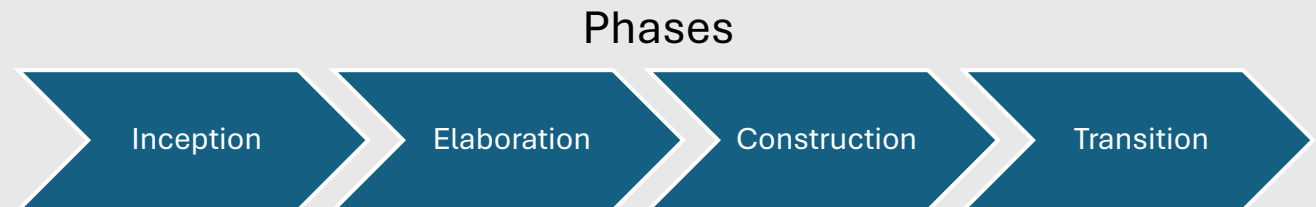
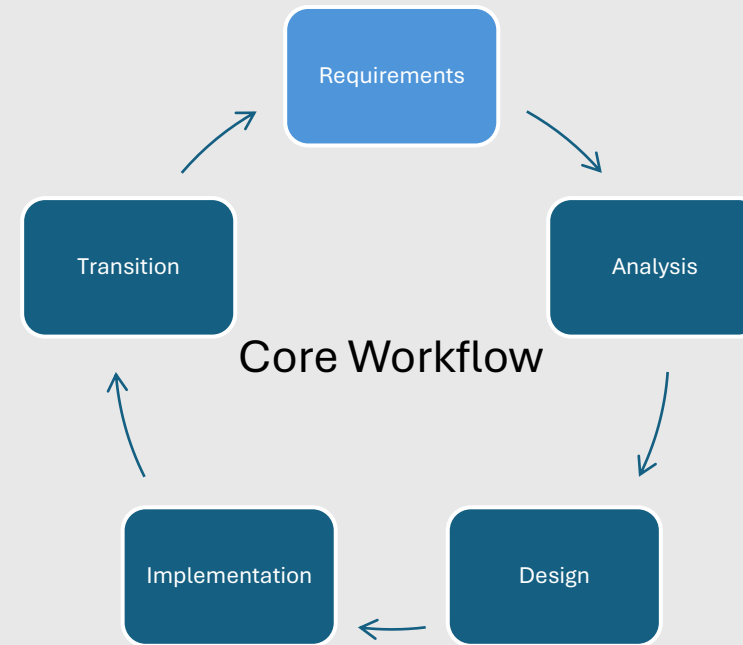
Iterations and increments

Architecture/Model-centric

Use case driven

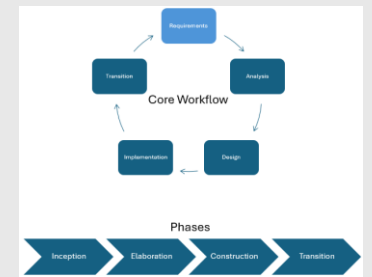
Not Agile

Unbalanced (see next slide)



Core workflows are iterated in each phase!



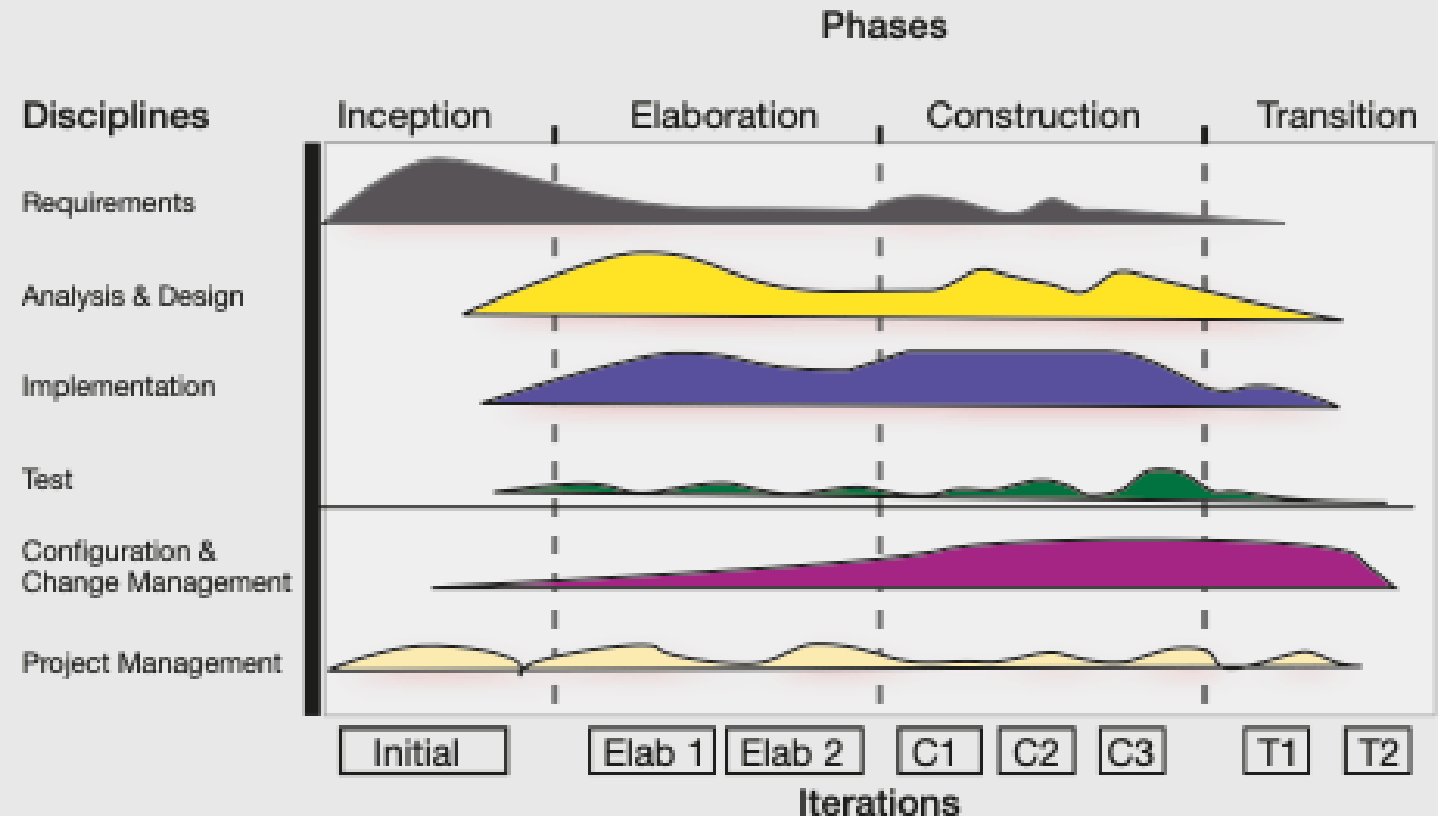


# Unified Process

**Inception** – define project & product scope, a vision and establish the business case

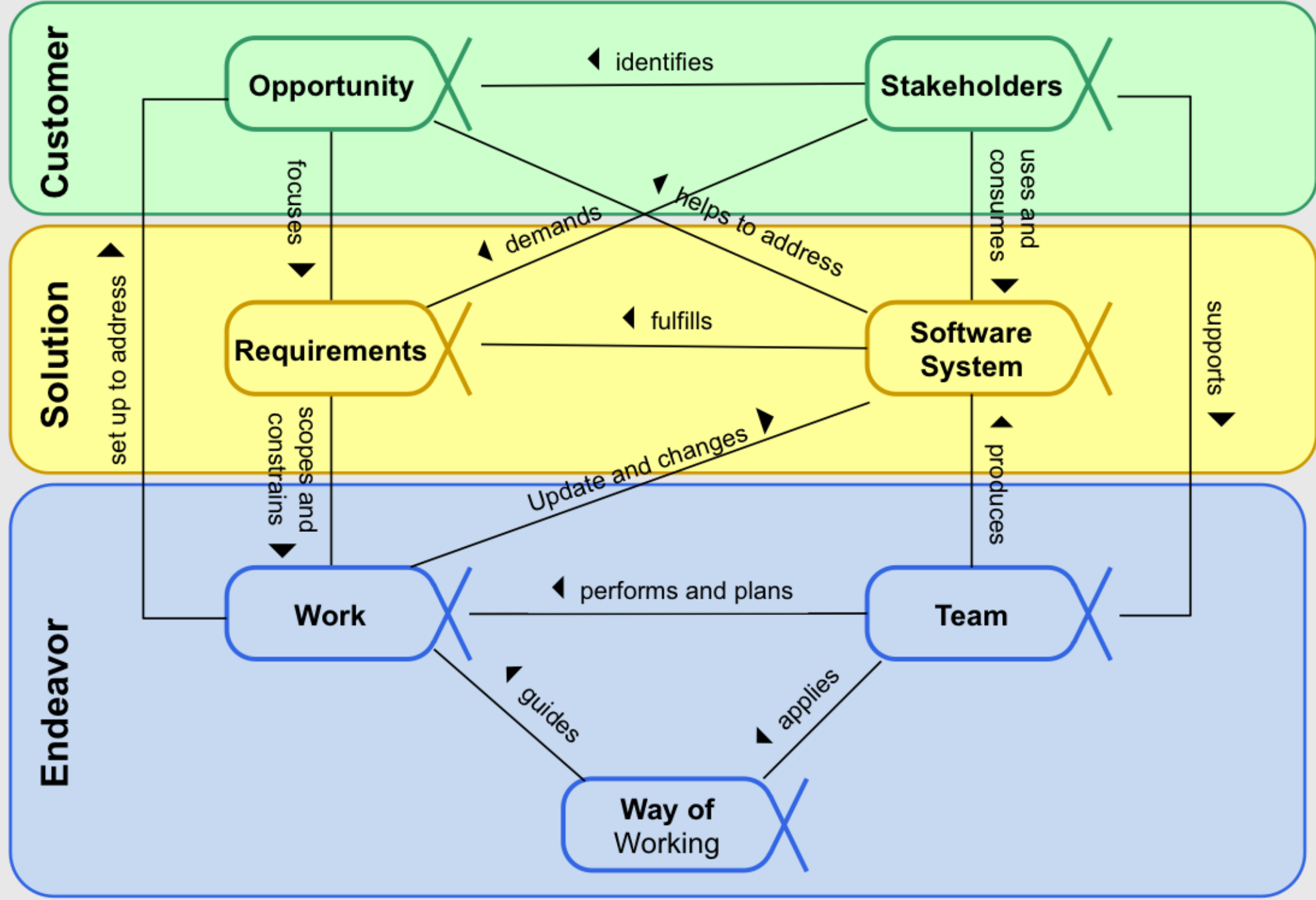
**Elaboration** – business and use case analysis. domain modeling, slowly shifting to design modeling, architecture centric

**Construction** – less analysis, more design, focus on achieving feature completeness according to construction plan



# SEMAT - Essence

- “Essence provides us with a thinking framework that allows teams to understand where they are and acts as a foundation for their way-of-working. At the heart of the Essence approach is the Essence Kernel – a simple state-driven model of software engineering.”





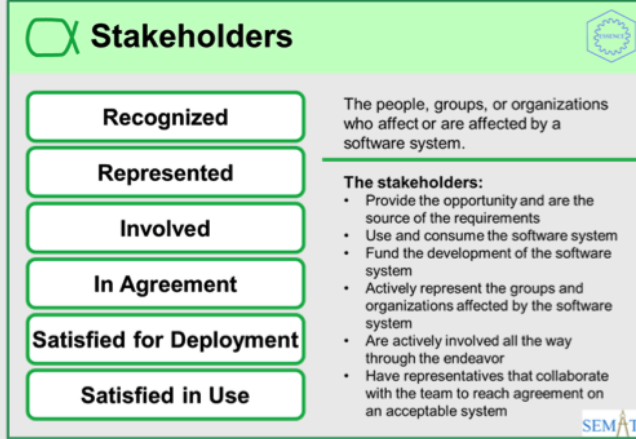
Overview Card



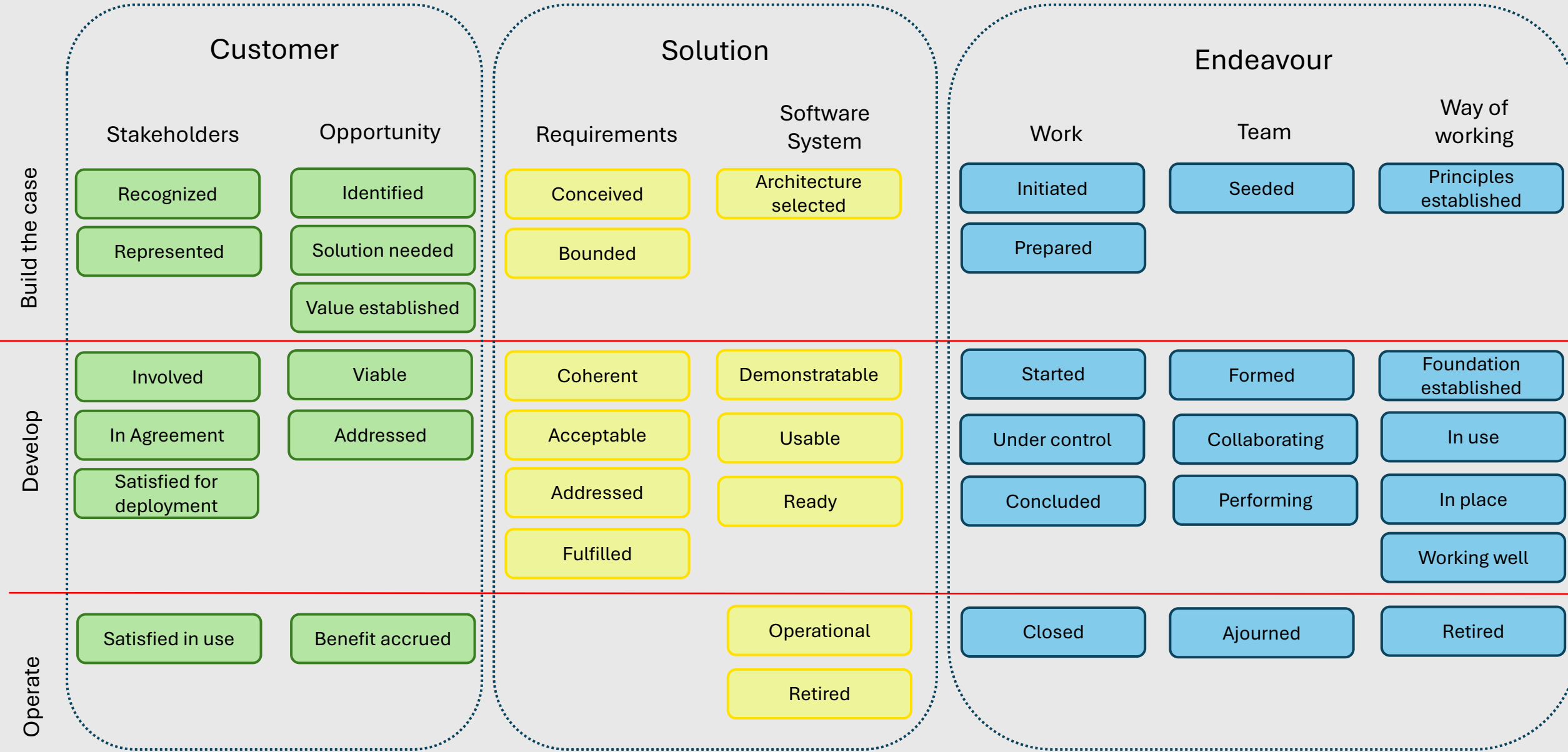
State Cards

| Recognized: Stakeholders have been identified                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Stakeholder groups identified</li> <li>Key stakeholder groups represented</li> <li>Responsibilities defined</li> </ul>                                                                                            | <p>All the different groups of stakeholders that are, or will be, affected by the development and operation of the software system are identified.</p> <p>There is agreement on the stakeholder groups to be represented. At a minimum, the stakeholder groups that fund, use, support, and maintain the system have been considered. The responsibilities of the stakeholder representatives have been defined.</p>                                                                                                                                                                       |
| Represented: The mechanisms for involving the stakeholders are agreed and the stakeholder representatives have been appointed                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <ul style="list-style-type: none"> <li>Representatives assigned</li> <li>Representatives authorized</li> <li>Collaboration approach agreed</li> <li>Way of working supported &amp; requested</li> </ul>                                                  | <p>The stakeholder representatives have agreed to take on their responsibilities. The stakeholder representatives are authorized to carry out their responsibilities. The collaboration approach among the stakeholder representatives has been agreed. The stakeholder representatives support and request the team's way of working.</p>                                                                                                                                                                                                                                                 |
| Involved: The stakeholder representatives are actively involved in the work and fulfilling their responsibilities                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <ul style="list-style-type: none"> <li>Representatives assist the team</li> <li>Timely feedback and decisions provided</li> <li>Changes promptly communicated</li> </ul>                                                                                 | <p>The stakeholder representatives assist the team in accordance with their responsibilities. The stakeholder representatives provide feedback and take part in decision making in a timely manner. The stakeholder representatives promptly communicate changes that are relevant for their stakeholder groups.</p>                                                                                                                                                                                                                                                                       |
| In Agreement: The stakeholder representatives are in agreement                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <ul style="list-style-type: none"> <li>Minimal expectations agreed</li> <li>Rep's happy with their involvement</li> <li>Rep's input valued</li> <li>Team's input valued &amp; requested</li> <li>Priorities clear &amp; perspectives balanced</li> </ul> | <p>The stakeholder representatives have agreed upon their minimal expectations for the next deployment of the new system. The stakeholder representatives are happy with their involvement in the work. The stakeholder representatives agree that their input is valued by the team and treated with respect. The team members agree that their input is valued by the stakeholder representatives and treated with respect. The stakeholder representatives agree with how their different priorities and perspectives are being balanced to provide a clear direction for the team.</p> |
| Satisfied for Deployment: The minimal expectations of the stakeholder representatives have been achieved                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <ul style="list-style-type: none"> <li>Stakeholder feedback provided</li> <li>System ready for deployment</li> </ul>                                                                                                                                     | <p>The stakeholder representatives provide feedback on the system from their stakeholder group perspectives. The stakeholder representatives confirm that they agree that the system is ready for deployment.</p>                                                                                                                                                                                                                                                                                                                                                                          |
| Satisfied in Use: The system has met or exceeds the minimal stakeholder expectations                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <ul style="list-style-type: none"> <li>Feedback on system use available</li> <li>System meets expectations</li> </ul>                                                                                                                                    | <p>Stakeholders are using the new system and providing feedback on their experience. The stakeholders confirm that the new system meets their expectations.</p>                                                                                                                                                                                                                                                                                                                                                                                                                            |

Full Checklist



# Example Go/No-Go tollgates



People

# People + Process = Success!?



What's the problem? Mostly people.

## **What I've learned the hard way:**

Communicating is difficult.

Listen, don't talk.

Everything is a gray scale.

A coin has two sides.

We're all different.

## **I asked my colleague Tobias C what he has learnt over the years:**

Sometimes there are cultural differences.

Be curious and ask.

Most people mean well, but some have sharp elbows and don't think about your best interests.

Some say one thing but mean another

There are team players, and then there are those who are not.

It takes a long time to make decision

Planting ideas is a good way to get the team on board



# More people problems

- Hidden agendas.
  - Career strategies
  - Personal clashes
  - Ego and pride
  - Stress & burnout
  - Poor resolution skills
  - Different work styles
- Anxiety, depression, self-doubt
  - Knowledge/skill gaps
  - Lack of
    - Respect
    - Trust
    - Motivation
    - Empathy

# Attributions

- Images from
  - [www.pixabay.com](http://www.pixabay.com)
  - [www.unsplash.com](http://www.unsplash.com)
  - [www.freepik.com](http://www.freepik.com)
- [mountaingoatsoftware.com](http://mountaingoatsoftware.com)